

SULTAN ISMAIL, Malaysia

WMO: 486790

Lat: 1.6384N

Lon: 103.6660E

Elev: 135

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.3	71.7	68.1	103.8	79.7	69.6	109.2	78.3	15.8	84.1	14.3	84.5	1.5	340	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.5	93.2	81.2	91.8	80.5	91.4	80.3	83.3	90.7	82.5	89.5	81.8	88.6	6.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.6	88.8	80.6	159.9	88.4	80.0	156.3	87.3	47.3	91.6	46.6	89.4	45.8	88.8	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.4	9.8	DB	69.7	95.0	1.7	1.5	68.5	96.1	67.5	96.9	66.5	97.7	65.3	98.8	98.5
(5)			WB	68.3	84.9	1.6	2.1	67.2	86.4	66.2	87.6	65.3	88.8	64.1	90.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	80.3	81.2	82.2	82.7	82.6	82.0	81.4	81.2	81.2	81.7	81.3	80.7
	DBStd	1.97	2.22	1.76	1.94	1.70	1.73	1.83	1.76	1.81	1.73	1.67	1.78	2.07
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11508	939	872	997	980	1010	960	972	967	937	982	940	952
	CDD65	6033	474	452	532	530	545	510	507	502	487	517	490	487
	CDH74	52829	3776	3944	4713	4743	5169	4865	4629	4519	4195	4459	3939	3878
	CDH80	19283	1269	1507	1877	1834	1936	1808	1690	1655	1543	1624	1304	1236
(14) Wind	WSAvg	3.3	4.6	4.7	3.5	2.7	2.6	2.7	3.0	3.1	2.9	2.8	2.9	3.8
(15) Precipitation	PrecAvg	101.6	9.3	5.5	8.3	9.6	7.8	6.9	6.7	7.9	7.7	9.2	10.8	11.8
	PrecMax	132.3	24.1	15.8	18.4	15.2	11.7	10.7	10.7	13.0	10.6	13.3	15.0	32.9
	PrecMin	81.7	2.0	1.9	3.0	5.7	3.4	4.4	1.4	4.6	2.4	5.6	4.4	5.2
	PrecStd	13.7	5.7	3.4	4.3	2.8	2.7	2.0	2.3	2.6	1.9	2.4	2.8	6.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	93.1	93.6	94.8	93.5	93.1	91.8	91.8	92.2	93.0	91.8	91.7
		MCWB	81.3	79.6	80.7	83.1	81.7	81.5	80.9	80.0	80.4	81.1	81.0	81.4
	2%	DB	89.8	91.3	92.3	92.9	91.7	91.5	91.0	91.1	91.2	91.3	90.0	89.9
		MCWB	79.6	79.2	80.2	81.5	80.9	80.6	80.3	79.8	80.1	80.1	80.0	80.1
	5%	DB	88.1	89.7	91.2	91.2	91.0	89.9	89.6	89.6	89.6	89.7	89.3	88.1
		MCWB	78.7	78.5	79.7	80.5	80.6	80.0	79.6	79.3	79.6	79.3	79.8	79.7
	10%	DB	86.4	88.0	89.5	89.5	89.3	88.2	88.0	87.9	87.9	88.0	87.5	86.3
		MCWB	77.9	78.1	79.0	80.1	80.4	79.9	79.4	79.1	79.5	79.4	79.6	78.9
	0.4%	WB	82.5	82.4	83.6	84.5	84.1	83.4	83.1	82.0	82.5	82.8	82.5	82.7
		MCDB	90.3	90.5	91.8	93.0	91.4	90.4	89.8	88.9	89.5	90.6	90.2	90.1
	2%	WB	81.0	80.9	81.9	83.0	82.7	82.2	81.6	81.0	81.4	81.4	81.4	81.3
		MCDB	88.1	88.5	89.9	90.4	89.1	88.6	88.1	87.7	88.2	88.7	88.5	88.1
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	79.7	79.8	80.9	81.8	81.8	81.3	80.6	80.2	80.5	80.5	80.4	80.1
		MCDB	86.0	86.9	88.5	88.6	87.9	87.5	87.2	86.6	87.1	87.3	86.9	86.3
	10%	WB	78.7	78.9	79.8	80.6	80.9	80.3	79.7	79.5	79.6	79.6	79.5	79.1
		MCDB	84.4	85.6	86.6	87.1	87.0	86.3	85.7	85.5	85.8	85.8	85.1	84.7
		MDBR	13.2	15.2	15.7	15.6	14.5	13.8	13.8	14.0	14.4	14.5	13.8	12.8
		MCDBR	15.6	17.2	17.8	17.4	16.4	15.7	15.7	16.0	16.3	16.3	15.8	15.2
(36) Mean Daily Temperature Range	5% DB	MCWBR	6.7	6.8	7.0	7.4	6.9	6.7	6.7	6.6	7.1	6.9	6.7	6.9
		MCDBR	14.3	15.4	16.1	16.0	14.9	14.4	14.2	14.6	14.9	14.9	14.6	14.3
	5% WB	MCWBR	6.9	7.1	7.5	7.9	7.3	7.1	6.9	6.9	7.3	7.1	6.9	7.0
(40) Clear-Sky Solar Irradiance	taub		0.466	0.479	0.491	0.494	0.485	0.498	0.489	0.530	0.559	0.582	0.480	0.464
	taud		2.292	2.234	2.209	2.200	2.234	2.198	2.235	2.098	2.015	1.930	2.249	2.303
	Ebn at Noon		274	272	268	261	256	249	253	248	246	242	268	272
	Edh at Noon		43	47	48	47	44	45	44	52	57	63	45	43
(44) All-Sky Solar Radiation	RadAvg		1422	1645	1643	1606	1492	1422	1407	1426	1418	1425	1357	1307
	RadStd		122	149	148	98	65	99	96	75	101	103	77	97

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.21	+0.86	+2.03	+1.08	+2.54	+2.29	N/A	N/A	+448
(47) Regional (7 neighbors)		+0.65	+0.18	N/A	N/A	+0.32	+0.31	N/A	N/A	+445

Nomenclature: See separate page